

Project Objectives and Requirements

PROCTECH 2SE3 – Systems Engineering Fundamentals

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Project Objectives

Functional Performance

1. The system should accept a mixed input batch of standard Canadian coinage (5¢, 10¢, 25¢, \$1, \$2) without manual pre-sorting.
2. The system should accurately sort Canadian coin denominations.
3. The system should process coins with high efficiency.
4. The system should calculate the total monetary value of the processed batch in real-time.
5. The system must maintain an individual count for each coin denomination (e.g., “14 Quarters, 5 Loonies”).
6. The system should prevent foreign objects (non-coins, washers, damaged coins, etc.) from entering the coin storage, sorting them into a dedicated “Reject Tray” to prevent contamination of valid bins.

System Reliability & Safety

7. The system should detect mechanical jams promptly.
8. Upon detecting a jam, the system shall automatically halt the intake motor to prevent motor burnout or coin damage.
9. The system should attempt an automated “clearance sequence” (e.g., reverse motor direction for 2 seconds) before signalling for operator intervention.
10. The system will retain the current count and value data in non-volatile memory to prevent data loss during a power outage.
11. The system must operate at a noise level suitable for commercial environments.

User Interface & Operational Feedback

12. The system must provide a “Session Reset” function that clears the current count for the next user.
13. The display should show the “Total Value” and “Breakdown by Coin” on the main screen upon completion of the cycle.
14. The system will detect when the hopper is empty (idle state) and automatically stop the sorting mechanism after a 5-second timeout.
15. The system should display a visual “Maintenance Required” alert if internal diagnostics report a degradation in sensor precision.
16. The system must allow the operator to manually pause the operation at any time safely via a physical button (E-Stop).

Capacity & Maintenance

17. The system should monitor the fill level of each individual storage bin (Nickel, Dime, Quarter, Loonie, Toonie).
18. The system will alert the operator when any specific bin approaches capacity.
19. If a bin reaches 100% capacity, the system will automatically re-route that specific coin type to an overflow storage area or pause intake.
20. The system should track the total lifetime count of coins processed to schedule routine maintenance (e.g., "Clean after 10,000 coins").
21. The internal storage containers must be designed for quick removal and emptying by the operator without tools.
22. The system should perform a "Self-Health Check" on startup to verify the sensor functionality before allowing coins to be fed.

Pairwise Comparison Chart

Objectives	Accuracy	Safety	Reliability	Capacity	SCORE
Accuracy	***	1	1	1	3
Safety	0	***	1	1	2
Reliability	0	0	***	1	1
Capacity	0	0	0	***	1

Accuracy

How accurately the system can perform its functions. Accuracy was given a score of 3 as the primary function of the system is to be able to correctly identify, count, and sort Canadian coins. Incorrect sorting or counting directly defeats the purpose of the system, despite speed or capacity.

Safety

The safety of the system for any users and operators. Safety was given a score of 2 since the protection of the operator and system must be ensured. Features such as fault detection, emergency stops, and enclosed components are essential for safe operation in commercial environments.

Reliability

Reliability was given a score of 1, as in order for long-term use, consistent performance is critical. The system must be able to preserve data during power loss, detect jams, and operate without frequent failure. If the system is reliable, that minimizes downtime and maintenance requirements.

Operational Efficiency

How efficiently the system executes its functions. Operational Efficiency was given a score of 1, as it must not come at the expense of accuracy, safety, or reliability. It can only be met after all other core objectives are satisfied.

Requirements

Functional

1. The system shall accept mixed standard Canadian coin denominations (5¢, 10¢, 25¢, \$1, \$2) without manual pre-sorting.
2. The system shall correctly identify and sort valid Canadian coins into their respective denomination bins.
3. The system shall physically reject non-coin objects and invalid items (e.g., foreign coins, washers, damaged coins) into a dedicated reject tray.
4. The system shall calculate and store the total monetary value of coins processed during each operating session.

Performance

5. The system shall process coins at a minimum rate of 60 coins per minute under normal operating conditions.
6. The system shall achieve a sorting accuracy of at least 99.5% for valid Canadian coin denominations.
7. The system shall detect and reject non-coin objects with at least 95% reliability.
8. The system shall complete the identification, sorting, and counting of each coin within 0.5 seconds of detection.
9. The system shall operate continuously for a minimum of 8 hours without performance degradation or system failure.

Constraint

10. The system shall stop accepting coins within 0.5 seconds of a mechanical jam condition.

11. The system shall preserve the current coin count and total monetary value data in the event of a power loss.
12. The system shall prevent operator access to internal components during normal operation.
13. The system shall include an emergency stop mechanism located within arm's reach of the operator.
14. The system shall operate using standard 120 V AC power supply in accordance with Canadian electrical safety standards.

Interface

15. The system provides a coin input hopper/chute interface that allows the operator to pour mixed Canadian coins into the system without tools.
16. The system shall provide separate physical output interfaces such as bins for each denomination (5¢, 10¢, 25¢, \$1, \$2).
17. The system shall provide a dedicated reject tray interface for non-coin objects and invalid coins.
18. The system shall provide a user interface that displays total value and per-denomination counts for the current session and indicates system status including Ready, Running, Jam, Fault.
19. The system shall provide a visual indicator interface, such as LEDs, to notify the operator of system states including Power On, Processing, Jam, and Fault.

Safety

20. The system shall provide a visual, and audible alarm to alert the operator when a fault or jam condition is detected.
21. The system shall ensure that all moving mechanical components are fully enclosed during normal operation to prevent accidental contact.
22. The system shall automatically shut down all moving components if an access panel or safety cover is opened during operation.
23. The system shall prevent operation unless all safety guards and enclosures are properly secured.

Requirements Verification

Test

All functional and performance requirements will be verified through controlled testing of the system using representative Canadian coin samples.

Inspection

Interfaces, electrical connections, physical components, and safety enclosures will be checked through visual and physical inspection.

Analysis

Calculations, and data handling will be checked by reviewing the system design and results.

Demonstration

System operation and safety features will be checked by demonstrating normal operation and fault conditions.

This report was authored by ALL members of PROCTECH 2SE3 Group 9, with minimal assistance from Artificial Intelligence (Google's 'Gemini 3 Pro' language model, February 2026), for the sole purpose of polishing language and improving clarity. All ideas, data analysis, and conclusions presented in this report are entirely our own.